

[Aim: 100/100 in Maths]

आरंभ

QUADRATIC EQUATIONS

Lecture #6

#ATK²:

Q.E

Factorisation

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

Nature of Roots

$$D = b^2 - 4ac$$

- $D > 0 \rightarrow$ Real & distinct
- $D = 0 \rightarrow$ Real & equal
- $D < 0 \rightarrow$ Unreal roots

WORD PROBLEMS

last chapter

PLEQV

Q linear Equations

this chapter

Quadratic Equation

1 Q.E

Word Problems:

#LP: John and Jivani together have 45 marbles. Both of them lost 5 marbles each, and the product of the number of marbles they now have is 124. Form the quadratic equation to find how many marbles they to start with, if John had x marbles. And hence, find the number of marbels each have?

$$\text{John} + \text{Jivani} = 45$$

Let John's marble $\Rightarrow (x)$ marbels $\xrightarrow{\text{lost}} (x-5)$ marbels
Jivani's marble $\Rightarrow (45-x)$ marbel $\xrightarrow{\text{lost}} (45-x-5) = (40-x)$ marbels

Ans $(x-5)(40-x) = 124$

$$\Rightarrow 40x - x^2 - 200 + 5x = 124$$

$$\Rightarrow 45x - x^2 - 200 + 124 = 0$$

$$\Rightarrow -x^2 + 45x - 324 = 0$$

$$\Rightarrow -(x^2 - 45x + 324) = 0$$

Quad

$$x^2 - 45x + 324 = 0$$

$$\Rightarrow x^2 - 36x - 9x + 324 = 0$$



$$p + q = -45$$

$$pq = 324$$

2	324
2	162
3	81
3	27
3	9
3	3

#LP: The sum of two numbers is 34. If 3 is subtracted from one number and 2 is added to another, the product of these two numbers becomes 260. Find the numbers. (CBSE 2022)

Let one no. is $(x) \longrightarrow (x-3)$

then, other is $\rightarrow (34-x) \longrightarrow (34-x+2) \Rightarrow (36-x)$

Ans $(x-3)(36-x)=260$

$$\Rightarrow \underline{36x} - \underline{x^2} - 108 + \underline{3x} = 260$$

$$\Rightarrow -x^2 + 39x - 108 - 260 = 0$$

$$\Rightarrow -x^2 + 39x - 368 = 0$$

$$\Rightarrow -(x^2 - 39x + 368) = 0$$

$$x^2 - 39x + 368 = 0$$

#LP: The product of two consecutive positive integers is 306. Form the quadratic equation to find the integers, if x denotes the smaller integer.

let x ¹, ²
 $(x+1)$

∴ two no.s are (x) & $(x+1)$

Ans

$$\Rightarrow (x)(x+1) = 306$$

$$\Rightarrow x^2 + x - 306 = 0$$



#LP: The product of two successive integral multiples of 5 is 300. Determine the multiples.

Let nos (x) , $(x+5)$

5 , 10



Ans

$$(x)(x+5) = 300$$

$$x^2 + 5x - 300 = 0$$

↓

two consecutive odd integers

3, 5
let no. are (x) , $(x+2)$



#LP: The denominator of a fraction is one more than twice the numerator. If the sum of the fraction and its reciprocal is $2\frac{16}{21}$ found, the fraction. (CBSE 2024)

Ans let num $\Rightarrow (x)$
den $\Rightarrow (1 + 2x)$ } $\rightarrow$ fraction $\Rightarrow \left[\frac{x}{2x+1} \right]$

ATQ $\Rightarrow \left(\frac{x}{2x+1} \right) + \left(\frac{2x+1}{x} \right) = 2\frac{16}{21}$

$$\Rightarrow \frac{x^2 + (2x+1)^2}{(2x+1)(x)} = \frac{58}{21}$$

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K3B

0.5

$$= \frac{1}{2}$$

>

$$\frac{1}{4}$$

$$= 0.25$$

$$\frac{1}{8}$$

>

$$\frac{1}{16}$$

$$\frac{1}{24}$$

>

$$\frac{1}{72}$$

$$\frac{1}{x}$$

>

$$\frac{1}{x+5}$$

den $\uparrow \Rightarrow$ total
value $\downarrow$

Time, Speed and Distance : $\Rightarrow D = S \times t \Rightarrow t = \frac{d}{s}$

#LP: A train, travelling at a uniform speed for 360 km, would have taken 48 minutes less to travel the same distance if its speed were 5 km/hr more. Find the original speed of the train.

$$\text{Distance (d)} = 360 \text{ km}$$

$$\text{let original speed} \Rightarrow x \text{ km/hr.}$$

$$\therefore \text{Original time } (t_1) = \left(\frac{360}{x} \right)$$

$$t_1 > t_2$$

ATQ

$$\text{New speed} \Rightarrow (x + 5)$$

$$t_2 = \left(\frac{360}{x + 5} \right)$$

$$\text{Difference of time} \Rightarrow \left(\frac{48}{60} \right) \text{ hr}$$

$$t_1 - t_2 = \frac{48}{60}$$

$$\frac{360}{x} - \frac{360}{x + 5} = \frac{48}{60}$$

$$90 \cancel{360} \left(\frac{1}{x} - \frac{1}{x+5} \right) = \frac{\cancel{48} \cancel{16} \cancel{x}}{\cancel{60} \cancel{20} 5}$$

$$\Rightarrow \frac{\cancel{x} + 5 - \cancel{x}}{x(x+5)} = \frac{1}{90 \times 5}$$

$$\Rightarrow (5)(90 \times 5) = x^2 + 5x$$

$$\Rightarrow 2250 = x^2 + 5x$$

$$\boxed{x^2 + 5x - 2250 = 0}$$

$\swarrow$
 $-45x + 50x$

✓

बस - दोली

$$\begin{array}{r} 5 - 2 = +3 \quad \swarrow +ve \\ \hline 2 - 5 = -3 \end{array}$$

#LP: In a flight of 2800 km, an aircraft was slowed down due to bad weather. Its average speed is reduced by 100 km/h, and by doing so, the time of flight is increased by 30 minutes. Find the original duration of the flight. (CBSE 2024)

$$D = 2800 \text{ km}$$

Let original speed $\Rightarrow x \text{ km/hr.}$

$$\text{Original time } (t_1) = \frac{2800}{x} \text{ hr}$$

$$t_1 < t_2$$

ATQ

$$\xrightarrow{2800 \text{ km}}$$

$$\xrightarrow{(x - 100)}$$

$$t_2 = \frac{2800}{x - 100}$$

$$\text{Diff} \Rightarrow \frac{30}{60} \text{ hr}$$

$$\cancel{t_1 = t_2}$$

$$t_2 - t_1 = \frac{1}{2}$$

$$\frac{2800}{x - 100} - \frac{2800}{x} = \frac{1}{2}$$

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THANK YOU COODIESS !!

